Khwaja Yunus Ali University

Lab Report -07

Name of the Department: Computer Science and Engineering

Course Code: CSE 0713-1105

Course Title: Electrical Circuit Lab

Experiment No.: 07

Name of the Experiment : Maximum Power transfer theorem using hardware and digital simulation .

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Instructor Signature & Date

Submitted by –	**Submitted to –**
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Experiment name : Maximum Power transfer theorem using hardware and digital simulation

Objective: The objectives of this experiment are:

- To verify the Maximum Power Transfer Theorem both practically and through simulation.
- To demonstrate that maximum power is delivered to the load when the load resistance equals the Thevenin resistance of the circuit.
- To simulate the circuit in Proteus software and compare the results with theoretical and practical calculations.
- To observe and analyze how power transfer varies with different load resistances.

Theory : According to the Maximum Power Transfer Theorem, the power transferred from a source to a load is maximized when the resistance of the load is equal to the internal (Thevenin) resistance of the source network. The power delivered to the load is given by:

$$P = \frac{V^2 R_L}{(R_{th} + R_L)^2}$$

Here,

- P: Power transferred to the load (in watts)
- V : Applied voltage across the network (in volts)
- R_{th} : Thevenin equivalent resistance (in ohms)
- R_L : Load resistance (in ohms)

Circuit Diagram :

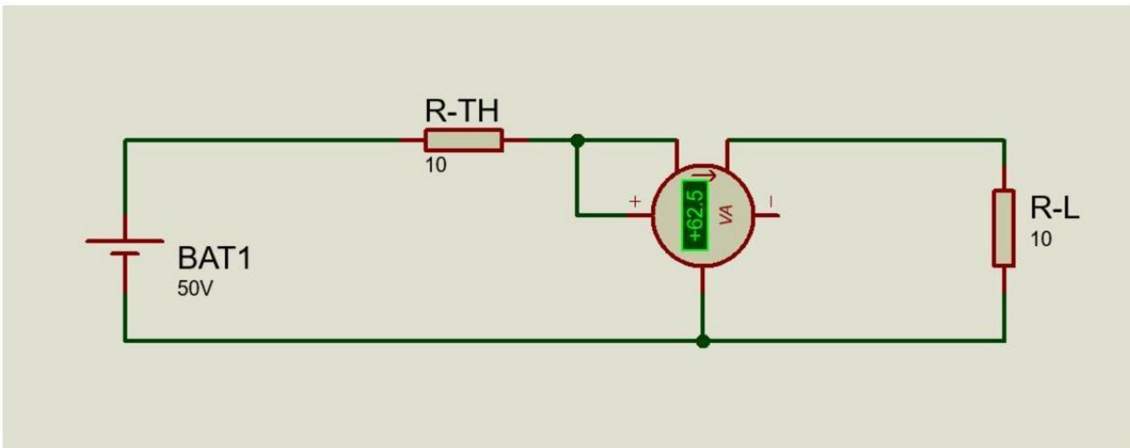

Figure 1: Simulation Circuit Diagram in Proteus

Apparatus Used :

- DC Voltage Source (50V)
- Fixed Resistor (10Ω) - Representing Thevenin Resistance
- Variable Load Resistors (5Ω, 8Ω, 10Ω, 12Ω, 15Ω)
- Wattmeter - For measuring power delivered to the load
- Digital Multimeter (optional) - For measuring voltage and current
- Proteus Simulation Software
- Connecting Wires and Breadboard (for hardware setup)

Procedure :

Hardware

1. Connect a 50 V DC power supply in series with a fixed resistor of 10Ω.
2. Connect a variable resistor as the load R_L.
3. Attach a wattmeter across the load to measure power.

4. Change the load resistance to $5\Omega, 8\Omega, 10\Omega, 12\Omega$, and 15Ω.
5. Record the wattmeter readings for each load.
6. Calculate the theoretical power using the formula.

Simulation (Proteus)

1. Open Proteus and build the same circuit as described above.
2. Use virtual instruments (DC source, resistors, wattmeter).
3. Simulate the circuit for each value of R_L.
4. Record measured power from the wattmeter.
5. Compare measured values with theoretically calculated ones.

Calculations :

The power transferred to the load for each resistance is calculated using the formula:

$$P = \frac{V^2 R_L}{(R_{th} + R_L)^2}$$

Given that $V = 50V$ and $R_{th} = 10\Omega$, we calculate for each R_L :

- For $R_L = 5\Omega$: $P = \frac{50^2 \times 5}{(10+5)^2} = 55.5$ W
- For $R_L = 8\Omega$: $P = \frac{50^2 \times 8}{(10+8)^2} = 61.6$ W
- For $R_L = 10\Omega$: $P = \frac{50^2 \times 10}{(10+10)^2} = 62.5$ W
- For $R_L = 12\Omega$: $P = \frac{50^2 \times 12}{(10+12)^2} = 62.0$ W
- For $R_L = 15\Omega$: $P = \frac{50^2 \times 15}{(10+15)^2} = 60.0$ W

Result Table :

S/N	Voltage (V)	Thevenin Resistance R_{TH}	Load Resistance $R_L(\Omega)$	Wattmeter Reading (W)	Calculated Power (W)
1	50	10	5	55.5	55.5
2	50	10	8	61.7	61.6
3	50	10	10	62.5	62.5
4	50	10	12	62.0	62.0
5	50	10	15	60.0	60.0

Table 1: Comparison of Measured and Calculated Power Transfer

Discussion :

The experimental and simulated results are consistent with the theoretical prediction of the Maximum Power Transfer Theorem. It is observed that:

- The power delivered to the load increases as R_L approaches R_{th}.
- The maximum power of 12.5 W occurs at $R_L = R_{th} = 10\Omega$.
- Beyond this point (for $R_L > R_{th}$), the power begins to decrease again.

This behavior verifies that for any given internal resistance in a network, maximum power is transferred only when the load resistance matches that internal resistance. This concept is critical in designing communication circuits, amplifiers, and power delivery systems.

Conclusion :

The experiment successfully demonstrated the Maximum Power Transfer Theorem using both hardware and Proteus simulation. It was clearly observed that the power transferred to the load is maximized when the load resistance equals the Thevenin resistance of the circuit. The theoretical calculations closely matched the practical and simulated results.

Precautions :

- Ensure that the power supply remains constant at 50 V throughout the experiment.
- Use properly rated resistors to avoid damage or inaccuracy due to overheating.
- Confirm all connections before powering the circuit to avoid short circuits.
- In Proteus, use ideal instruments to minimize simulation error.
- Record readings accurately after the system reaches steady state.